1 Documentation edition and implementation scope

This is the 5.3.0 documentation edition, dated 2026-11-09. It carries forward the system and service contracts from the retained 5.2.2 documentation and identifies the current primary implementation sources. The former `documentation/v5.2.2` folder is retained unchanged.

Document version 5.3.0 is distinct from executable, dependency, registered system/service and wire-contract versions. Values still defined by the source (including 5.2.2 artefact versions, 1.0.0 service versions and `SysMonitor-AMQP-IDD-5.2.2` payload identifiers) are preserved. This documentation update does not change runtime behaviour or establish a new implementation release. Historical verification statements describe the tests recorded in the inherited baseline; this revision validates document compilation and references.

2 localCloudManagement HTTP interface

The actual listener is plain HTTP; the dedicated launcher defaults to `http://127.0.0.1:8480/`. The inherited registry descriptor uses generic_http/policy NONE. Do not label this as a native HTTPS/mTLS interface. An external proxy and its certificates are outside this source relocation, and cannot repair legacy provider authentication by themselves.

2.1 Actual handler inventory

Method	Path	Behavior
GET	`/health`	Text OK; not dependency readiness
GET	`/` and `/ui`	Embedded dashboard; no console session
GET	`/app.js`, `/assets/app.js`	Embedded JavaScript
GET	`/styles.css`, `/assets/styles.css`	Embedded CSS
GET	`/api/session`	authenticated/configured state
POST	`/api/session`	Password login, success204with cookie
DELETE	`/api/session`	Logout,200with expired cookie
GET	`/api/overview`	Registry/runtime aggregation
GET	`/api/service-registry-management`	Devices/systems/services/runtime grouping
GET	`/api/datamanager`	Historian adapter; limit1..1000, default240
GET	`/api/consumers`	Inherited partial consumer view
GET	`/api/intercloud`	Static unavailable placeholder
GET	`/api/sysmonitor`	Inherited observation adapter, optional system query
GET	`/api/sysmonitor/subscriptions`	Read adapter, synthetic intake flags
PUT, DELETE	`/api/sysmonitor/subscriptions/{provider}`	Confirmed acknowledgement ONLY; no write
GET	`/api/plantdescriptions`	Plant/registry catalogue

Method	Path	Behavior
POST	`/api/plantdescriptions`	Confirmed provider create; currently200
PUT, PATCH	`/api/plantdescriptions/{id}`	Confirmed provider update; both forward PUT
DELETE	`/api/plantdescriptions/{id}`	Confirmed provider removal
GET	`/api/orchestration/rules`	Simple-store catalogue/rules
POST	`/api/orchestration/rules`	Confirmed validation then provider create
DELETE	`/api/orchestration/rules/{id}`	Confirmed provider removal
GET	`/api/logs`	Project-container logs, service/tail query
POST	`/api/containers/{service}/{action}`	Confirmed allowlisted project lifecycle
GET	`/localcloud/management/status`	Separate legacy requester status
GET	`/localcloud/management/systems`	Separate legacy management registry query
GET	`/localcloud/management/services`	Separate legacy management service query
GET	`/localcloud/monitor/echo`	Separate legacy echo

2.2 Session and access

All `/api/` requests require allowed Host/Origin host names and a configured password hash. Non-session routes also require a valid console cookie. Host-name matching ignores exact ports. Missing/invalid hash503, rejected Host/Origin403, absent/expired session401. Static UI/health and legacy routes are not covered by this console middleware.

POST session accepts JSON with password string. The configured hash is scrypt with N16384,r8,p1. A correct password returns204and sets `ah_console`, HttpOnly, SameSite=Strict, Path=/, Max-Age default28800. There is no Secure flag or port isolation. GET session returns authenticated/configured. DELETE removes the token and expires the cookie. Five wrong passwords in five minutes trigger one-minute lockout;429includes Retry-After60. Wrong password401.

Mutations require exactly `X-Arrowhead-Action:confirm`. Missing confirmation428with JSON error. Axum body/content-type extraction can reject malformed JSON before the handler checks confirmation. This header is not cryptographic replay protection.

2.3 Mutation DTOs

Rule creation requires consumer, serviceInstanceId and positive unsigned priority. Unknown/incompatible producer choice409; invalid required values400. The provider body wraps the candidate in a candidates array. Remove uses ids query at the inherited simple-store endpoint. Provider success is returned as JSON, generally200; provider/network failures502with error. No distributed transaction or atomic rollback is guaranteed.

Plant create/update JSON is forwarded to the inherited provider path. PATCH also forwards PUT. Create currently returns200. Delete forwards DELETE. Subscription PUT and DELETE ignore the body and return providerSystem/updated=true without applying any provider change; a200response is not configuration confirmation.

Lifecycle action must be start, stop or restart. A static source allowlist and Compose project filter constrain selection. Unknown action400, unlisted target403, unknown returned project service404, Docker failure502. Success JSON contains service/action/accepted=true/at and currently returns200, not202. The raw Docker socket privilege is not reduced by these application checks.

2.4 Read DTOs and errors

Overview includes collectedAt, cloud, summary, devices, systems, services, registryManagement, naming, infrastructure, orchestrations, containers, history, registration and errors. Some flags and collections are placeholders. Registry grouping uses devices, runtimeOnly and summary; no assurance of authoritative readiness follows from these fields.

Historian returns collectedAt/count/latest/entries/series; source data comes from inherited per-system/per-service SenML paths. Consumers uses TestCloud labels and empty authorized/orchestrated lists. Intercloud is static unavailable. SysMonitor reads group observations with sourceTime/value/quality, but upstream errors are hidden as empty data; subscription intake status is synthetic.

Logs accept service and tail (clamped1..500, default150). Response includes service/lines/collectedAt. Logs may include any container returned for the project, even if it is not lifecycle-allowed. Runtime rows include id/name/service/displayName/image/state/status/health/created/managed.

Console handler errors generally use a JSON error string. Legacy shared-runtime errors use errorMessage/errorCode/exceptionType/origin. Unknown routes404; unsupported methods405 through Axum. Exact Java DTO/error parity and a full browser deployment test are not established.

3 References and qualification

See `docs/source-manifest.json` for original/relocated source SHA256 records, `VERIFICATION.md` for checks actually run, and `docs/LocalCloudManager.sysml` for the descriptive, unparsed variability model. No Java interchangeability, native TLS or sysMonitor publisher is claimed. Existing cloud source and running deployment remain unchanged.

4 Revision record

2026-11-09: Documentation edition 5.3.0 derived from the retained 5.2.2 sources; metadata and local references updated. Runtime and wire-contract versions remain source-defined.